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**Kidney Tumor Conditional Generation via
Latent Diffusion**
Master's Thesis (15 ECTS)

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Kidney Tumor Conditional Generation via Latent Diffusion

Abstract:

Diagnostic imaging systems such as computed tomography (CT) are widely used in medical practice as it enables non-invasive visualization of internal body structures and play an important role in diagnosing diseases such as kidney cancer. However, identifying tumors in CT images is time- and cost-intensive because it requires specialized expertise by trained radiologists. To support this process, computer-aided diagnosis systems were introduced, but traditional approaches often relied on hand-crafted features and had limited generalizability. This has led to the adoption of deep learning-based methods, whose performance depends heavily on large amounts of annotated data. In medical imaging, such data is often difficult to obtain because of high annotation costs and privacy constraints. Synthetic data generation has therefore emerged as a promising alternative and has shown encouraging results in medical imaging. This thesis investigates whether synthetically generated kidney tumor CT images can reduce the need for annotated real data while maintaining competitive segmentation performance. To address this, the open-source MAISI framework was adapted for conditional kidney tumor generation by fine-tuning the ControlNet component, introducing inpainting, and modifying the loss to focus on the tumor region and its surrounding context. Alternative synthetic data generation methods were also included for comparison. The results show that the modified MAISI generally outperformed the alternative methods, especially when little or no real data were available. The best overall trade-off between segmentation quality and robustness to false positive predictions was achieved when MAISI-generated synthetic data were combined with 75% real data, corresponding to 120 CT scans. In this setting, performance was comparable to, or slightly better than, the real-data baseline, while false positive predictions were reduced. Overall, the findings suggest that synthetic kidney tumor CT data can serve as a useful supplement to real annotated data, particularly in low-data settings, but not yet as a full replacement.

Keywords: Latent Diffusion, Synthetic Data, Medical Imaging, Computed Tomography

CERCS: B110 Bioinformatics, medical informatics, biomathematics, biometrics; P176 Artificial intelligence

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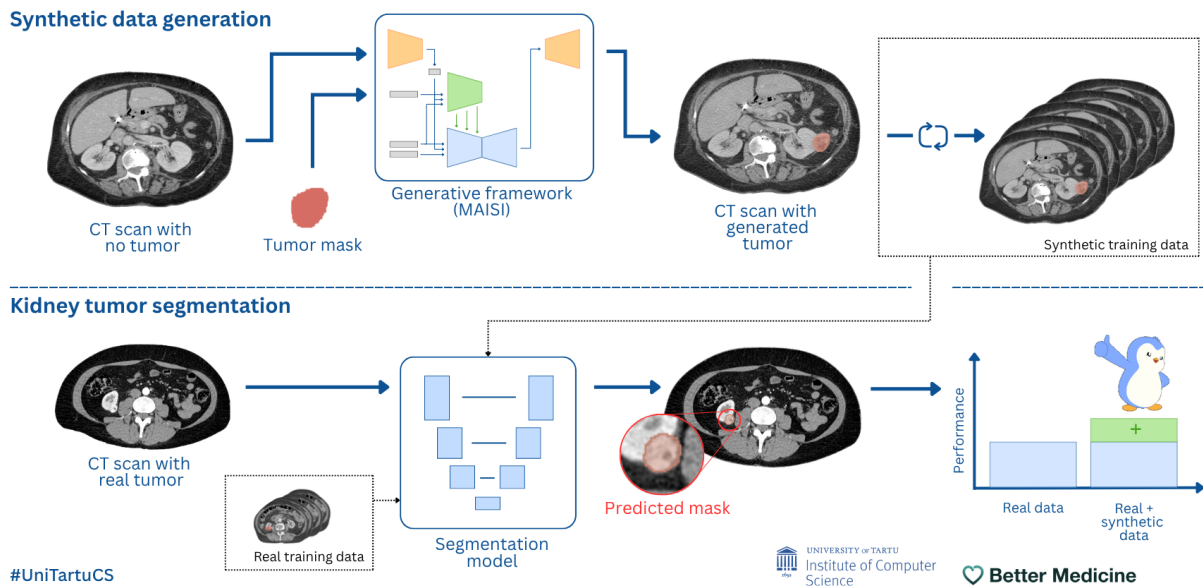


Figure 1. Graphical abstract.

Neerukasvaja tingimuslik genereerimine latentse difusiooni kaudu

Lühikokkuvõte:

Diagnostilisi kuvamissüsteeme, näiteks kompuutertomograafiat (KT), kasutatakse meditsiinipraktikas laialdaselt, kuna need võimaldavad keha sisemisi struktuure visualiseerida mitteinvasiivselt ning mängivad olulist rolli selliste haiguste nagu neeruvähi diagnoosimisel. Kasvajate tuvastamine KT-kujutistel on aga aja- ja ressursimahukas, sest see nõuab väljaõppinud radioloogide eriteadmisi. Selle protsessi toetamiseks võeti kasutusele arvutipõhised diagnostikasüsteemid, kuid traditsioonilised lähenemised tuginesid sageli käsitsi loodud tunnustele ning nende üldistusvõime oli piiratud. See on viinud sügavõppel põhinevate meetodite kasutuselevõtni, mille toimivus sõltub suurel määral suurest hulgast annoteeritud andmetest. Meditsiinilises kujutamises on selliseid andmeid sageli keeruline hankida kõrge annoteerimiskulu ja privaatsuspiirangute tõttu. Seetõttu on sünteetiliste andmete genereerimine kujunenud paljulubavaks alternatiiviks ning on näidanud julgustavaid tulemusi meditsiinilises kujutamises. Käesolevas töös uuritakse, kas sünteetiliselt genereeritud neerukasvaja KT-kujutised võivad vähendada annoteeritud reaalandmete vajadust, säilitades samal ajal konkurentsivõimelise segmenteerimistulemuse. Selleks kohandati avatud lähtekoodiga MAISI raamistikku tingimuslikuks neerukasvajate genereerimiseks, peenhäälestades ControlNet-komponenti, lisades inpainting-lähenemise ning muutes kaofunktsiooni nii, et see keskenduks kasvaja piirkonnale ja seda ümbritsevale kontekstile. Võrdluseks kaasati ka alternatiivseid sünteetiliste andmete genereerimise meetodeid. Tulemused näitavad, et kohandatud MAISI ületas üldiselt alternatiivseid meetodeid, eriti siis, kui reaalseid andmeid oli vähe või need puudusid üldse. Parim tasakaal segmenteerimiskvaliteedi ja valepositiivsete ennustuste suhte robustsuse vahel saavutati siis, kui MAISI abil genereeritud sünteetilised andmed kombineeriti 75% reaalandmetega, mis vastas 120-le KT-kujutisele. Selles seadistuses oli tulemus võrreldav või veidi parem kui ainult reaalandmetel põhinev baaslahendus, samal ajal kui valepositiivsete

ennustuste arv vähenes. Üldiselt viitavad tulemused sellele, et sünteetilised neerukasvaja KT-andmed võivad olla kasulikuks täienduseks annoteeritud reaalandmetele, eriti väheste andmete tingimustes, kuid ei kujuta veel endast nende täielikku asendust.

Võtmesõnad: Latentne Difusioon, Sünteetilised Andmed, Meditsiiniline Kujutamine, Kompuutertomograafia

CERCS: B110 Bioinformaatika, meditsiiniinformaatika, biomatemaatika, biomeetrika; P176 Tehisintellekt

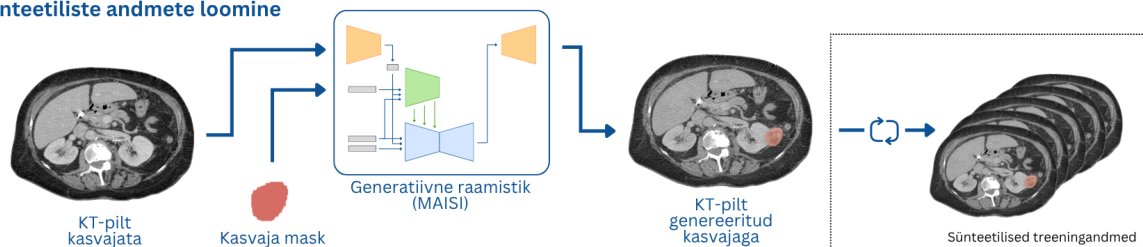
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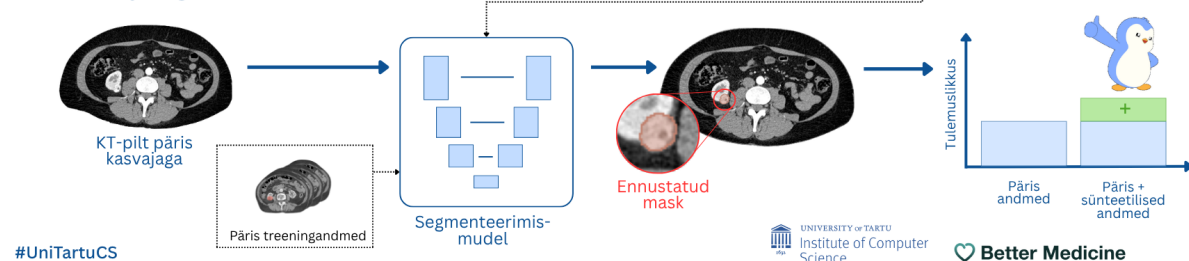
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Sünteetiliste andmete loomine



Neerukasvaja segmentatsioon



Joonis 2. Graafiline abstrakt.

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1. Introduction

Diagnostic imaging systems such as computed tomography (CT) play a vital role in medical practice by enabling the visualization of the body’s internal structures without the need for invasive procedures [1]. They are essential for the diagnosis and treatment planning of serious diseases such as kidney cancer, where early detection and timely initiation of treatment are critical, as identifying tumors before progression to advanced stages can be life-saving [2–4]. However, identifying tumors in CT images is both time- and cost-intensive, as it requires specialized expertise and is typically performed by trained radiologists with extensive clinical experience [5].

To alleviate the high cost, time demands, and reliance on expert interpretation in medical image analysis, computer-aided diagnosis (CAD) systems were introduced to assist radiologists in identifying abnormalities in medical images [6]. These systems are designed to improve diagnostic accuracy and reduce inter-observer variability by highlighting suspicious regions for further analysis [7]. However, traditional CAD approaches often rely on hand-crafted features and may struggle to generalize across diverse datasets, which has led to the adoption of deep learning-based methods [8].

Studies published in recent years have proposed AI-driven, or deep learning based, methods for the segmentation of lesions or organs in medical images, where segmentation refers to the task of assigning image regions to a specific organ or lesion, and such methods can potentially support clinical practice by helping to localize and delineate abnormalities more efficiently, which may contribute not only to earlier disease detection, but also to improved lesion characterization, more consistent reporting, and better treatment planning [9–11]. For example, Isensee et al. [9] developed a deep learning-based segmentation method, nnU-Net, that automatically configures itself — including data preprocessing, network architecture, training, and post-processing — for new biomedical segmentation tasks. In a recent work, Petryshak et al. [10] introduced a deep learning-based tool that uses a 3D U-Net architecture from the nnU-Net framework [9] to detect and characterize kidney cancer. A different approach from the aforementioned studies was introduced by Wang et al. [11], who adapted the Segment Anything Model (SAM) [12], originally developed for segmenting objects in natural images based on user-provided prompts, to 3D medical imaging by training the model from scratch on 82 diverse medical datasets. Their work highlights that model architectures and methodological advances alone may not be sufficient to achieve strong segmentation performance without access to large amounts of relevant data, such as CT scans containing annotated tumors.

Conversely, training on limited annotated datasets can substantially constrain the performance and generalizability of deep learning models [8, 13]. However, acquiring labeled medical data is both costly and time-consuming, as it requires annotation by trained specialists such as radiologists, who often face substantial workload and time constraints [14]. Given these challenges, simply increasing the volume of annotated data is not always feasible. Consequently, researchers have explored alternative strategies for obtaining labeled data, among which synthetic data generation has emerged as a promising approach [15–17].

Although this direction has already shown encouraging results in medical imaging, its potential to support kidney tumor segmentation in 3D CT imaging still requires further investigation. Therefore, the main research question of this thesis is whether synthetically generated kidney tumor CT images can reduce the need for large amounts of manually annotated real-world data while maintaining competitive segmentation performance. More specifically, this thesis investigates whether synthetic data generated using adapted deep learning-based generative

models can support kidney tumor segmentation at a level comparable to models trained on real data, and to what extent synthetic data can complement limited real datasets. In this work, kidney tumors are chosen as the focus because our lab already has access to relevant datasets and prior experience working with this type of lesion. Restricting the study to a single lesion type also makes the adaptation, training, and evaluation of the generative frameworks more feasible. Once such a pipeline is established for one target lesion, it can later be extended to additional lesion types.

To address this research question, it is first necessary to consider the main deep learning approaches that have been used for image synthesis. Early deep learning approaches for image generation, such as Generative Adversarial Networks (GANs), demonstrated that neural networks could learn the underlying distribution of image data and generate realistic synthetic images [18]. More recently, diffusion models have emerged as an alternative generative approach and have shown strong performance in image synthesis, including in medical imaging [19–21].

At the same time, generating high-resolution 3D medical images such as CT scans, which can have dimensions larger than 512^3 voxels, remains computationally demanding [22]. One way to reduce this cost is to perform generation in a lower-dimensional latent space, which can preserve important image information while making the process more efficient [23]. This approach has also been explored in medical imaging for the generation of synthetic data that may support downstream tasks such as segmentation [24].

In 2025, NVIDIA introduced one of the most prominent open-source frameworks — MAISI (Medical AI for Synthetic Imaging) — which applies latent diffusion models to three-dimensional medical image generation [25]. MAISI addresses two primary limitations in medical imaging: data scarcity and the high cost of human annotation. Its primary objective is to generate anatomically accurate 3D CT images, supporting image sizes up to $512 \times 512 \times 768$. The authors demonstrated that MAISI achieves comparable performance in segmentation tasks when trained on synthetic and real data, highlighting its potential to mitigate the challenges of limited data availability and expensive annotation.

MAISI can generate not only anatomical structures but also several types of lesions, including hepatic, lung, pancreatic, and colon tumors, as well as bone lesions [25]. However, since MAISI was neither trained for kidney tumor generation nor adapted for inpainting, it cannot be directly applied to conditional kidney tumor generation, which is the main objective of this thesis. To address these limitations, we introduce several modifications to the framework and make the following key contributions:

- We adapt the framework for an inpainting task, enabling the generation of organs and lesions at specified locations in CT images of healthy subjects based on a provided mask.
- We modify the loss function to focus on the region of interest and its surrounding context, rather than on the entire image.
- We fine-tune MAISI for the task of kidney tumor generation.
- We also conducted segmentation experiments to compare the performance of models trained on real data and on synthetic data generated using our modified version of MAISI and alternative approaches.

In the preparation of this thesis, ChatGPT-5 (Plus) by OpenAI was used for grammar checking, wording and sentence structure refinement, punctuation correction, and exploration of alternative

terminology to improve consistency and clarity throughout the thesis. It was also used occasionally to search for background information, and all sources suggested by the application were manually validated before inclusion in the thesis. ChatGPT was also used to assist in drafting Python scripts for the experiments; however, all generated code was reviewed, tested, and validated by the author before use in the research.

2. Background

This section introduces the main concepts and methods relevant to this thesis. First, we briefly discuss kidney cancer and the role of computed tomography as a central imaging modality in its diagnosis and treatment planning. We then provide the necessary background on medical image segmentation and synthetic image generation, including the generative frameworks most relevant to this work. Finally, we introduce the MAISI framework, which serves as the main basis for the methodological developments presented in this thesis.

2.1 Kidney cancer

This work focuses on a single lesion type, namely kidney tumors. This choice allows the synthetic data generation pipeline to be established and evaluated for one target lesion before extending it to additional lesion types in future work. Kidney tumors were selected because our lab has access to relevant datasets and prior experience working with this type of lesion. In general, kidney cancer is a disease in which malignant tumors develop in the kidneys [26]. It may not cause noticeable symptoms in its early stages and is therefore often diagnosed only after it has spread beyond the kidneys [27]. In many cases, kidney tumors are discovered incidentally during imaging performed for other reasons, although early detection of such lesions can be life-saving [28]. Since medical imaging plays a central role in the diagnosis and treatment planning, it is important to briefly introduce the main imaging modality used in this context. One of the primary imaging modalities used to diagnose kidney cancer, as well as many other cancers originating in the abdominal region, is computed tomography [29]. This modality is discussed in more detail in the following section.

2.2 Computed tomography

A CT image is produced using X-rays, which are a form of electromagnetic radiation capable of penetrating most materials, including biological tissue [30]. During a CT scan, X-rays are transmitted through the body while detectors positioned opposite the X-ray source measure the intensity of the transmitted radiation (see Figure 3), which is then processed using reconstruction algorithms to produce cross-sectional grayscale images (slices) that can be stacked to form a three-dimensional representation of the anatomy, with slice thickness, which determines the spacing between slices, typically ranging from 1 to 10 mm [31]. As X-ray radiation passes through the body, the difference between the emitted and detected intensities reflects tissue attenuation, which depends on the tissue’s density and physical properties, with denser materials such as bone blocking more radiation than water or soft tissue [32]. Attenuation values are expressed in Hounsfield Units (HU), named after Godfrey Newbold Hounsfield, who was awarded the Nobel Prize for his role in the development of computed tomography [33, 34]. The HU scale is defined using two reference materials: air, assigned a value of -1000 HU, and water, assigned a value of 0 HU [35]. For other materials, the Hounsfield Unit can be calculated using the following definition:

$$HU = 1000 \times \frac{\mu - \mu_{water}}{\mu_{water} - \mu_{air}},$$

where μ represents the linear attenuation coefficient of the material, μ_{water} is the attenuation coefficient of water, and μ_{air} is the attenuation coefficient of air [36].

However, since CT images typically span a wide range of HU values, from approximately -1000 HU to values exceeding 1000 HU, distinguishing tissues with similar attenuation can

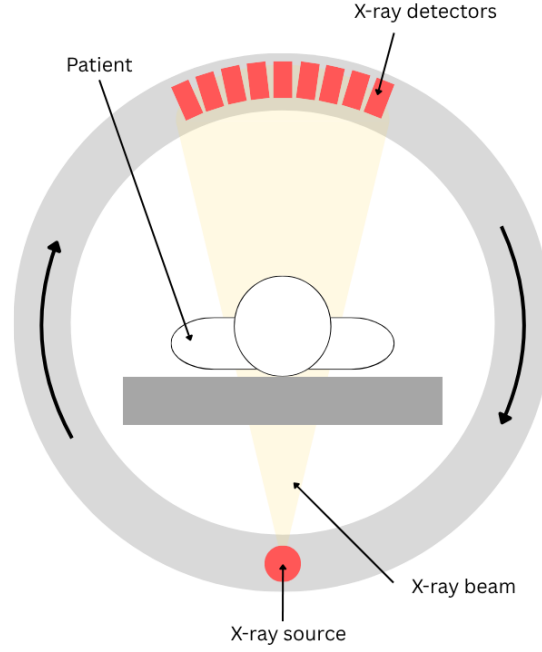


Figure 3. Schematic illustration of a CT scanner showing the rotating X-ray source and detectors around the patient.

be challenging when using the full intensity range [37]. To address this challenge, a technique called windowing can be applied, whereby a specific interval of HU values is selected and mapped to the available intensity range, with values below the interval set to the minimum intensity and values above it set to the maximum intensity [38]. This process is defined by two parameters: window level (WL) and window width (WW), where the window level specifies the center of the selected HU interval and the window width determines its size, from which the minimum and maximum values can be calculated using the following formula [39, 40]:

$$[WL - \frac{WW}{2}, WL + \frac{WW}{2}]$$

A typical soft-tissue window is defined by $WL = 40$ and $WW = 400$, corresponding to the HU interval $[-160, 240]$. This means that values below the lower bound are clipped to -160, whereas values above the upper bound are clipped to 240 [39, 41]. Given that this soft-tissue window is frequently used, the interval $[-160, 240]$ was adopted in this thesis for all experiments involving windowed CT data. Figure 4 illustrates the difference between an unwindowed and a windowed CT slice.

While windowing is used to improve the visibility of structures by adjusting the displayed intensity range, CT images also require careful handling of their spatial representation. Because CT is a three-dimensional imaging modality, scans can be viewed in three anatomical planes: axial, coronal, and sagittal. The axial plane, which is the most commonly used, corresponds to a view from the patient's feet (inferior) toward the head (superior), whereas the sagittal plane provides a side view that divides the body into left and right parts, and the coronal plane divides the body into the front (anterior) and back (posterior) parts of the body [43, 44] (see Figure 5). In addition, different CT scanners, file formats, and post-processing tools may rely on different coordinate conventions, which can result in differences in image orientation and spatial

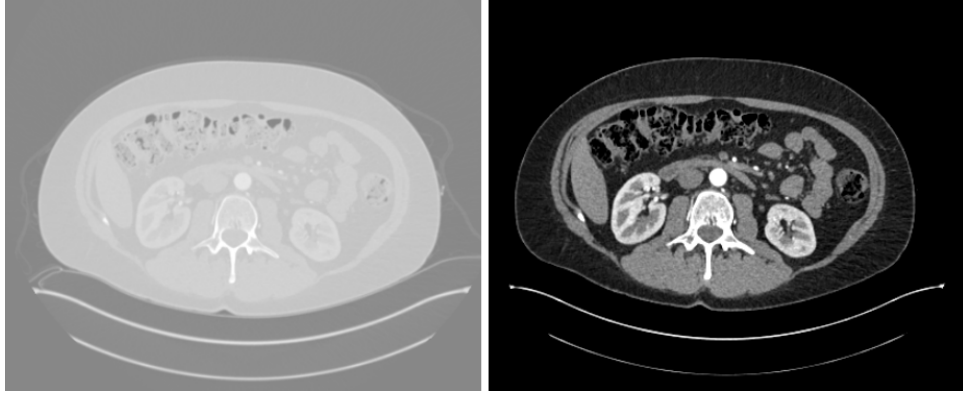


Figure 4. Example axial slice from a non-windowed CT scan (left) and a windowed CT scan (right) from the KiTS23 dataset [42].

interpretation after conversion or further processing [45, 46]. The most common coordinate systems are LPS and RAS [46]:

$$LPS = \begin{cases} \text{from right towards left} \\ \text{from anterior towards posterior} \\ \text{from inferior towards superior} \end{cases}$$

$$RAS = \begin{cases} \text{from left towards right} \\ \text{from posterior towards anterior} \\ \text{from inferior towards superior} \end{cases}$$

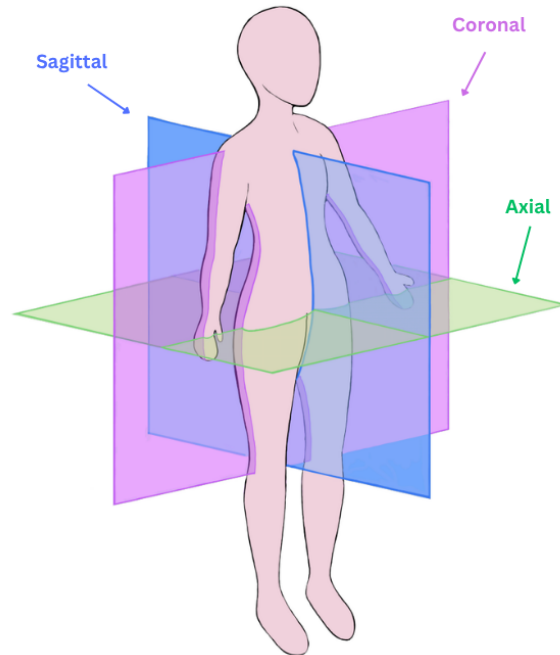


Figure 5. Illustration of the anatomical planes used in medical imaging: axial, sagittal, and coronal.

These spatial properties are important because CT scans are treated as three-dimensional volumes in medical image analysis, and differences in orientation, spacing, or alignment can affect how anatomical structures and their corresponding annotations are interpreted.

2.3 Medical image segmentation

Clinical tasks such as lesion detection, lesion characterization, diagnosis, and treatment planning often require anatomical structures or lesions to be accurately identified in medical images [10]. However, manually delineating these regions is time-consuming and requires expert knowledge [5]. One way to automate this process is to assign each pixel or voxel in the image a specific label, such as background, organ, or lesion. This process is known as segmentation and has become an important area of research in 3D medical image analysis.

In recent years, many medical image segmentation tasks have been addressed using deep learning models trained on annotated medical images [47]. One of the most widely used architectures in medical image segmentation is U-Net [48], originally developed for biomedical image segmentation. In this context, a model architecture refers to how a neural network is structured to take an input image and produce a segmentation result. U-Net became especially influential because it captures both fine image details and the broader surrounding context, which is important for accurate segmentation of organs and lesions [48].

However, strong segmentation performance depends not only on the network architecture itself, but also on many additional design choices. These include how the images are preprocessed, what patch size is used, how the training procedure is configured, and which post-processing steps are applied [9]. Since these choices often need to be adapted to the properties of each dataset, building an effective segmentation pipeline can require substantial manual effort [9].

To reduce this effort, nnU-Net was introduced as a self-configuring framework that automatically adapts key components of the segmentation pipeline to the properties of a given dataset [9]. This means that the framework automatically determines important settings related to preprocessing, network configuration, training, and post-processing instead of requiring them to be manually tuned for each new task. For example, manually selecting suitable training settings, such as the learning rate or patch size, can be time-consuming and often requires repeated experiments, whereas nnU-Net determines such settings based on the dataset properties. For this reason, nnU-Net was chosen for the experimental part of this thesis, as it reduces the need for manual tuning and provides a robust framework for comparing the effects of different training data settings. The preprocessing pipeline of nnU-Net [9] includes cropping to remove unnecessary image regions, resampling to standardize spacing across scans, and intensity normalization to make CT values more consistent. In addition, the framework applies several augmentation techniques during training, including random rotations, scaling, elastic deformations, gamma augmentation, and mirroring [9].

However, conventional data augmentation only transforms existing images and therefore remains limited by the variation already present in the dataset, such as differences in organ shape, tumor size, tumor appearance, and lesion location. This is because conventional augmentation methods usually modify the image globally or geometrically, rather than introducing genuinely new anatomical structures or lesion patterns. Synthetic data generation can therefore be viewed as a complementary way of expanding the training set by introducing additional examples with different tumor characteristics and spatial configurations, which may improve the robustness of the segmentation model [49, 50].

2.4 Medical image synthesis methods

Image synthesis has become an important area of research in medical imaging, where generative models are used to generate realistic images by learning anatomical patterns from training data [51]. Deep learning models such as Generative Adversarial Networks and diffusion models have demonstrated strong performance in medical image synthesis tasks [52, 53].

One of the first deep learning approaches capable of generating realistic images was the Generative Adversarial Network, introduced by Goodfellow et al. [18] in 2014, which marked a major breakthrough in generative modeling. GANs consist of two neural networks trained in competition with each other: a generator and a discriminator. The generator takes random noise as input and learns to produce synthetic samples that resemble the training data, while the discriminator receives both real and generated samples and learns to distinguish between them. Through this adversarial training process, the generator gradually improves by learning to create samples that are increasingly difficult for the discriminator to classify as fake. However, GANs have several disadvantages, including training complexity and instability caused by the need to balance the optimization of both networks [54].

Because of these limitations, more recent work has increasingly focused on alternative generative frameworks. In particular, diffusion models have emerged as a powerful alternative generative framework with more stable optimization behavior, among which one of the most influential modern approaches is the Denoising Diffusion Probabilistic Model (DDPM), introduced by Ho et al. [20]. DDPMs are generative models that learn to generate data through a sequential denoising process. Building on earlier work on diffusion probabilistic models (DPMs) [55], DDPMs use a Markov chain in which Gaussian noise is gradually added to the data distribution through a forward diffusion process, eventually transforming the input image into nearly pure noise. The model is then trained to learn the reverse process, which progressively denoises a noisy sample to recover a realistic data sample (see Figure 6). In this framework, a noisy image at timestep t is obtained as follows:

$$x_t(x_0, \epsilon) = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

where x_0 denotes the original image and $\bar{\alpha}_t$ controls the noise level.

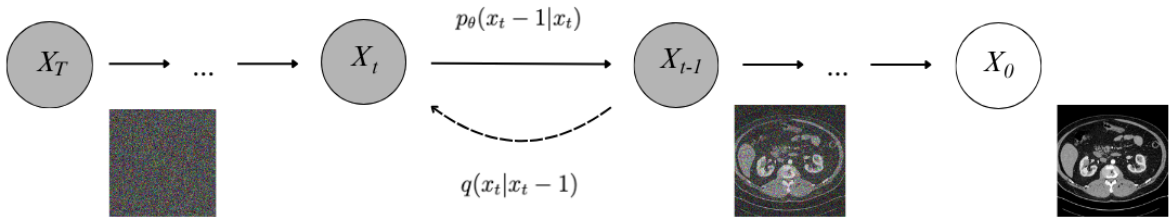


Figure 6. Forward and reverse processes in diffusion models. The forward process $q(x_t | x_{t-1})$ adds Gaussian noise to the input image over time, while the reverse process $p_\theta(x_{t-1} | x_t)$ learns to remove noise step by step, generating a realistic image from an initial noise sample [20].

However, diffusion models typically require substantial computational resources, particularly for high-resolution three-dimensional images such as CT scans, which can have dimensions larger than 512^3 voxels [22]. To mitigate this limitation, Rombach et al. [23] demonstrated that performing the diffusion process in a learned latent space — a lower dimensional representation

of the image that preserves essential structural and semantic information — can significantly reduce computational requirements without degrading image quality. For this purpose, different variants of autoencoders can be employed. An autoencoder is a neural network composed of two main parts: an encoder, which learns a compact representation of the input by extracting its most relevant features, and a decoder, which learns to reconstruct the original input from this compressed representation [56]. Within the latent diffusion framework of Rombach et al. [23], a pretrained perceptual autoencoder, trained to preserve important visual information, is used to encode images into latent space and decode the generated latent representations back into image space. In one reported comparison, their latent diffusion model required $4\times$ fewer training resources than the pixel-space Ablated Diffusion Model (ADM) introduced by Dhariwal and Nichol [57], while using half as many parameters. Furthermore, Khader et al. [24] showed that generating high-resolution medical images using latent diffusion models, and subsequently using the synthetic data for downstream segmentation tasks, can improve segmentation performance. While latent diffusion provides an efficient way to generate high-resolution 3D medical images, not all methods used in this thesis rely on latent-space generation. In particular, the 2D and 2.5D methods considered here perform synthesis directly in image space.

2D and 2.5D methods. In this thesis, we use synthetic data generated using 2D and 2.5D methods developed by the co-supervisor of this work. In these approaches, the author utilizes a Denoising Diffusion Probabilistic Model, specifically the Palette framework [58], adapted to better handle medical imaging data. The model architecture in these methods is based on a U-Net [48] with attention layers, which helps preserve fine details and maintain contextual consistency. Unlike latent diffusion approaches, these methods perform data synthesis directly in image space, allowing the model to learn the data distribution without potential information loss introduced by latent compression.

In the 2D approach, individual slices taken from a 3D volume are processed independently. Each slice containing the tumor mask is handled separately, and synthetic tumor textures are generated on a per-slice basis. This approach enables lightweight training, fast iterations, and a large number of slice-wise training examples. However, it lacks volumetric context, which may lead to artifacts and inconsistencies in tumor appearance across adjacent slices.

The 2.5D approach incorporates information from neighboring slices by stacking adjacent slices as input, thereby capturing limited three-dimensional contextual information. This improves anatomical consistency across slices and enhances the realism of generated tumors, particularly along the z-axis, while maintaining relatively low computational cost. However, the limited receptive field may still result in inconsistencies when modeling larger lesions.

DiffTumor. Another method used in this thesis for synthetic data generation is the DiffTumor framework, developed by Chen et al. [17]. In contrast to the 2D and 2.5D approaches, DiffTumor is a latent diffusion framework in which the authors adapted the Vector Quantized Generative Adversarial Network (VQGAN) architecture [59], replacing 2D convolutions with their 3D counterparts in order to compress 3D CT images. Within the DiffTumor pipeline, the diffusion model is conditioned on both tumor masks and healthy regions of CT volumes. This means that the model is trained to generate tumors based on masks that specify their size and location, rather than generating the entire image from scratch. The tumor masks used for generation are also created within the framework itself. DiffTumor supports the generation of three tumor types: liver, pancreatic, and kidney tumors. By combining the generated tumor masks with healthy CT images, DiffTumor uses them to expand the training data available for segmentation models.

TS-Radiomics. In addition to the approaches described above, this thesis also explores a generative framework called TS-Radiomics, proposed by Kim et al. [60]. The main idea of this approach is to use radiomic features to guide tumor mask and tumor generation [60]. The authors use PyRadiomics [61], an open-source platform for extracting radiomic features from medical images. Radiomic features are quantitative descriptors derived from medical images, including, for example, features related to tumor shape, texture, and intensity statistics such as mean, maximum, minimum, variance, and percentiles [62]. In the TS-Radiomics framework, PyRadiomics is used to extract shape, texture, and intensity-based features from a user-provided 3D CT or MRI image together with a corresponding tumor mask [60]. These extracted features are then used for tumor mask and tumor generation [60].

For tumor mask generation, the framework uses GigaGAN [63] with cross-attention, conditioned on the extracted shape features [60]. Similar to DiffTumor, TS-Radiomics also uses VQGAN [59] to compress 3D images and perform the diffusion process in latent space [60]. However, instead of DDPM, TS-Radiomics uses the Brownian Bridge Diffusion Model (BBDM) [64], in which generation is formulated as a transformation from a reference image to a target-style image while preserving the information that should remain unchanged; in the case of TS-Radiomics, this reference image corresponds to a healthy CT or MRI scan [60, 64]. For tumor generation, the framework uses the generated mask, the compressed 3D healthy CT or MRI image, and the extracted texture and intensity-based features as conditioning inputs [60]. TS-Radiomics also supports healthy tissue generation through tumor removal when radiomic features are not provided [60]. The authors evaluate the framework on tumors in four organs, namely the kidney, lung, brain, and breast, across both CT and MRI data, demonstrating its potential for medical data augmentation [60].

However, based on our experiments, the TS-Radiomics framework was not able to handle large CT images exceeding a size of 128^3 , even on an H200 GPU with 141 GB of memory. This image size is not suitable for our segmentation experiments, in which we aim to compare CT volumes of size 512^3 to ensure that both kidneys and the tumors are fully included in each image. Based on our prior experience, segmentation performance may degrade when the organs are not fully present. For this reason, we did not pursue TS-Radiomics further for synthetic data generation in this work.

2.5 Medical AI for Synthetic Imaging (MAISI)

Among the synthetic data generation approaches considered in this thesis, we chose to focus more closely on the MAISI framework. MAISI is an open-source framework developed by NVIDIA and consists of three main components: a volume compression network, used to compress 3D volumes into a compact latent space; a latent diffusion model, which generates the full CT volume based on specified configurations; and ControlNet, which injects conditional mask information into the latent diffusion model [25]. Unlike the DiffTumor and TS-Radiomics methods discussed above, it includes foundation models such as a volume compression network and a latent diffusion model, trained on 37,243 and 10,277 CT volumes, respectively [25]. As a result, it provides a strong backbone for adapting the framework to new generation tasks for which it was not originally trained, such as kidney tumor generation, without the need to train a generative model from scratch on large amounts of data, since the pretrained framework has already learned a broad prior over 3D CT anatomy and image structure. Additionally, MAISI has been trained to generate five types of lesions — liver tumor, lung tumor, pancreatic tumor, colon tumor, and bone lesion — making it easier to extend the focus from kidney tumors to other organs in the future without needing to train the model for each task [25]. In this thesis,

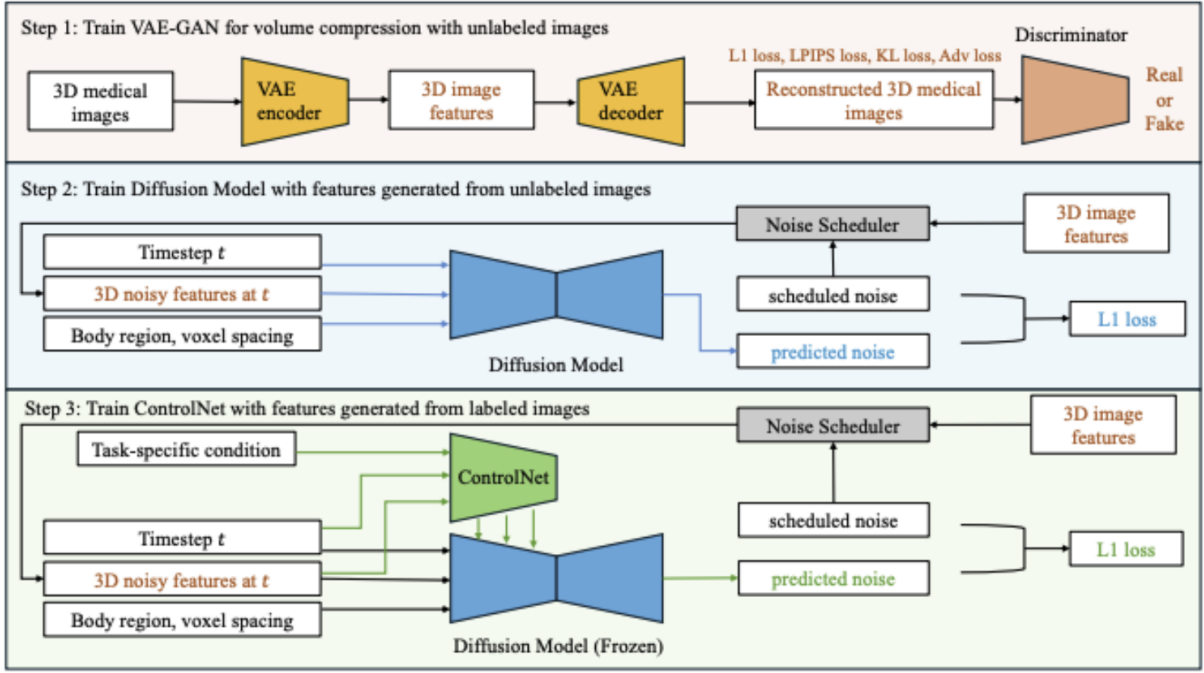


Figure 7. Overview of the three-stage MAISI training pipeline [25]. Stage 1 illustrates the training of a VAE-GAN, in which 3D CT images are encoded into latent representations by the VAE encoder and then reconstructed by the VAE decoder. The reconstructed images are evaluated by the discriminator, which learns to distinguish between the original 3D images and the reconstructions [25, 67]. Stage 2 shows the training of the latent diffusion model, in which noise is added to the latent image features at a given timestep t , and the model learns to predict this noise conditioned on imaging metadata such as body region and voxel spacing [23, 25]. Stage 3 depicts the integration of ControlNet, which processes task-specific conditions (e.g., tumor masks) and injects the resulting feature maps into the latent diffusion model to enable conditional image generation [25, 68].

we use an enhanced version of the framework, MAISI-v2, introduced by Zhao et al. [65] in 2025, where the authors additionally support rectified flow [66] as an alternative to the standard DDPM-based diffusion process, enabling an approximately $33\times$ faster diffusion process while preserving image quality. Figure 7 provides an overview of the three development stages of MAISI.

2.5.1 Main components

Volume compression network. As mentioned earlier, MAISI performs the diffusion process in a latent space [25]. For this purpose, the authors implemented a variational autoencoder combined with a discriminator from the GAN architecture to compress a 3D CT image into a latent representation [25]. This combination, known as VAE-GAN, was originally proposed by Larsen et al. [69]. During VAE-GAN training, the input image is encoded by the VAE encoder and reconstructed by the decoder. The discriminator is then trained to distinguish real images from reconstructed ones, while its intermediate feature representations are used to measure how similar the reconstruction is to the original image [69]. In addition to adopting VAE-GAN for the volume compression network, MAISI also addresses one of the main limitations related to GPU memory requirements, since processing a large 3D volume, such as 512^3 , can quickly exceed hardware limits [25]. To mitigate this, the framework introduces a feature-map splitting strategy in which the latent representation is divided into overlapping segments that can be processed

separately. Each segment can be assigned to a different GPU or processed sequentially on a single GPU, enabling the generation of high-resolution 3D volumes while reducing memory requirements [25].

Rectified flow. After compressing the 3D volume into a latent representation, the next step is the diffusion process. In MAISI-v2, the authors implemented Rectified Flow, originally proposed by Liu et al. [66], alongside DDPM. The main difference between these two approaches is that DDPM gradually denoises an image through many stochastic steps, whereas Rectified Flow seeks a more direct path from the source distribution to the target distribution by modeling the transformation as an ordinary differential equation (ODE). As a result, Rectified Flow requires fewer inference steps and can generate images more efficiently, making the process faster [65, 66].

ControlNet. The MAISI generation pipeline also incorporates ControlNet, a neural network architecture designed to add additional conditioning signals to a diffusion model while preserving the behavior of the original pretrained model [25, 68]. Fine-tuning a large diffusion model, such as the one used in MAISI, is computationally expensive due to its high memory requirements, and when only a small dataset is available, it may also lead to overfitting or the loss of previously learned knowledge [68, 70]. To address this, MAISI adopts ControlNet, which allows the diffusion model to incorporate additional conditioning information, such as tumor masks, so that the generated structures follow the specified anatomical guidance more closely without requiring the foundation model to be retrained [25].

2.5.2 Original generation pipeline

Based on inspection of the implementation and the conducted experiments, the MAISI framework generates an entire 3D volume conditioned on the provided masks. When multiple candidate masks are available for a given organ or lesion, the framework randomly selects one during inference for the anatomies specified in the configuration file. Based on the implementation, organ masks appear to be used largely as provided, whereas lesion masks undergo additional processing and augmentation. In particular, the framework removes the lesion from its original location, applies transformations to the mask, such as rotation and resizing, constrains the lesion to remain within the corresponding organ region, verifies that the deformed lesion retains at least 80% of its original size, smooths the mask boundaries, and places the lesion at a new location.

Masks are not the only required inputs for inference. It is also necessary to specify the body region, such as head, chest, or abdomen; the anatomy list, that is, the organs or lesions to be generated; the number of inference steps for both volume and mask generation; and the autoencoder sliding-window inference size and overlap. The last two parameters control the sliding-window inference used during reconstruction. Based on our experiments, generation quality appears to depend on sliding-window inference size and overlap: larger values generally improve output quality, but also increase computational cost. In addition, the output size and voxel spacing of the generated volume must be specified.

Based on the source code available in the official GitHub repository ¹, there are limitations on the size of the generated 3D volumes. For width and height, the valid sizes are 256, 384, and

¹ <https://github.com/Project-MONAI/tutorials/blob/main/generation/maisi>

512, whereas valid depths are 128, 256, 384, 512, 640, and 768. Voxel spacing is not explicitly constrained; however, the first two spatial dimensions must have equal spacing.

During inference, pure noise is sampled with the specified latent shape, which was empirically observed to have a downsampling factor of 4 in each spatial dimension relative to the final output. The noisy latent, together with the provided mask and timestep t , is passed through ControlNet, which produces feature maps for each residual block and for the middle block of the latent diffusion model. These feature maps are then injected into the model, which outputs denoised latent features at timestep t . After all timesteps are completed, the VAE-GAN decoder reconstructs the final 3D volume from the generated latent representation. The final inference output consists of a synthetic 3D volume and the corresponding segmentation mask containing labels for the organs and lesions specified in the anatomy list. The labels used in both the input masks and the masks generated by MAISI are predefined by the authors of the framework. Some of these labels serve as placeholders and can be reassigned when adapting the framework to new generation tasks. In our case, placeholder label 129 was used for kidney tumor, while the predefined labels 5 and 14 were used for the right and left kidney, respectively.

3. Data and methods

In this section, we provide an overview of the datasets used in this work and describe how the data was preprocessed for both synthetic data generation and downstream segmentation experiments. We also explain how MAISI was adapted for inpainting and fine-tuned for kidney tumor generation. In addition, since alternative approaches were also used for synthetic data generation and subsequent segmentation experiments, some modifications to DiffTumor and TS-Radiomics were required. However, due to limitations of the TS-Radiomics framework, it could not ultimately be used in our experiments, although we still attempted to adapt it to our task. All computationally demanding stages of synthetic data generation and downstream segmentation were performed on the Tartu University high-performance computing (HPC) cluster².

3.1 Datasets

In this work, we used the publicly available KiTS23 dataset, created for the 2023 Kidney and Kidney Tumor Segmentation Challenge [42]. The dataset annotations include three classes — kidney, tumor, and cyst — and its publicly available training set contains 489 tumor-containing CT images. In this thesis, KiTS23 was used as one of the main sources of tumor-containing CT scans for fine-tuning MAISI for kidney tumor generation. Another publicly available dataset, which includes the same three classes as KiTS23 was KIRC³, from which we used 180 tumor-containing CT images. In addition, we used a proprietary dataset provided by Better Medicine [10], denoted here as TUH, containing 191 tumor-containing CT images. Together, these datasets complemented KiTS23 during fine-tuning, while the TUH dataset also provided the real tumor-containing scans used in the downstream segmentation experiments.

We further included an additional 160 healthy CT scans also provided by Better Medicine, referred to in this work as control scans. Together with their corresponding tumor masks, created and provided by the co-supervisor, these scans served as the common input for all synthetic data generation methods. In this way, the healthy control scans and their masks formed the basis of the synthetic datasets used in this thesis, providing the anatomical background and spatial guidance needed to insert tumors at predefined locations. In the following section, we describe how the datasets were preprocessed and how they were incorporated into the different stages of the generation pipeline, including MAISI fine-tuning, and synthetic data generation.

3.2 Development of the synthetic data generation pipeline

The synthetic data generation pipeline began with fine-tuning MAISI for kidney tumor generation. For this purpose, we used the full KiTS23 training set together with the TUH and KIRC datasets. From the TUH dataset containing 191 scans, 181 were used for MAISI training, while the remaining 10 were set aside for an initial validation of generation quality. In total, this provided 850 tumor-containing CT images for fine-tuning MAISI for kidney tumor generation. Since CT scans can vary considerably in characteristics such as size, voxel spacing, orientation, and range of HU values, especially when they are obtained from different sources, several preprocessing steps were required to standardize the datasets before they could be used to train the generative models. Without such preprocessing, the resulting variability could introduce

² <https://hpc.ut.ee/>

³ <https://portal.gdc.cancer.gov/projects/TCGA-KIRC>

unwanted inconsistencies that could make training more difficult and potentially lead to poorer generations.

3.2.1 Data preprocessing

Before training the generative models, several preprocessing steps were applied to standardize the CT data. For fine-tuning MAISI, the tumor-containing CT images and their corresponding segmentation masks were cropped or padded around the tumor to a size of $512 \times 512 \times 256$. The orientation was converted to RAS and the voxel spacing was standardized to $1 \times 1 \times 1$ mm. In addition, the segmentation masks were relabeled to match the MAISI label convention, including separate labels for the left and right kidney and the MAISI-specific kidney tumor label. For fine-tuning DiffTumor, the same preprocessed CT images were used, while the masks were converted back to the original label convention. After completing these preprocessing steps, we first proceeded with the MAISI fine-tuning pipeline.

3.2.2 Fine-tuning MAISI for kidney tumor generation

As described in Section 2.5, the MAISI framework was not originally trained for kidney tumor generation and therefore could not be used directly for our task. In this thesis, fine-tuning MAISI for kidney tumor generation referred specifically to fine-tuning the ControlNet component, which injects conditional mask information into the generation process. Although the term *training* was occasionally used for simplicity, the procedure described here corresponded more precisely to fine-tuning, since it built on the pretrained MAISI framework rather than training the model from scratch. Before this stage, we first analyzed the framework architecture and pipeline to identify which components required modification. The fine-tuning pipeline consisted of two main stages. In the first stage, the training data (850 CT scans) were encoded into latent representations, referred to here as embeddings, since the diffusion process in MAISI operates in latent space. In the second stage, the training script was run on the encoded data and the corresponding segmentation masks using adjusted configuration files.

As mentioned, before fine-tuning, the 850 training CT scans were encoded into latent representations using the pretrained MAISI autoencoder. Although MAISI uses a splitting strategy to reduce memory usage, encoding images of size $512 \times 512 \times 256$ still required substantial computational resources. On average, creating one embedding took approximately 4-5 minutes, and processing the entire training set (850 CT scans) took about 3 days, after which the pipeline could proceed to the second stage, namely fine-tuning the ControlNet component.

The fine-tuning stage used the embeddings created in the previous step together with their corresponding segmentation masks. For the final version of the fine-tuned ControlNet, we used a batch size of 1, a learning rate of 1×10^{-5} , and 2000 training epochs. A batch size of 1 was required because increasing it led to out-of-memory errors. These settings were selected based on a series of preliminary experiments. During development, we explored learning rates ranging from 1×10^{-6} to 1×10^{-3} and numbers of epochs between 100 and 10,000. Initial experiments with fewer epochs, between 100 and 500, produced tumors with very poor visual quality, lacking both texture and contrast (see Figure 8). To better understand whether the model was capable of learning the target task at all, we deliberately overfit the ControlNet on a single training scan and examined whether it could learn to reproduce the exact tumor. These experiments showed that the number of training epochs had to be increased substantially, from hundreds to thousands. When we trained the ControlNet for 10,000 epochs on a single scan, the model was eventually able to generate a tumor that was very similar to the one in that scan (see Figure 9), and the same behavior was then confirmed on several other scans. Based on these findings, we returned to training on the full dataset and selected 2000 epochs as a reasonable compromise. The earlier

experiments showed that a few hundred epochs were clearly insufficient, whereas 10,000 epochs would have required an impractical amount of time on the full training set. In practice, training for 2000 epochs still took approximately 2 weeks, but a visual inspection indicated that it could produce tumors with a plausible appearance.



Figure 8. Example of a CT image with a tumor generated by MAISI after fine-tuning for only a few hundred epochs. The tumor region is outlined in red.

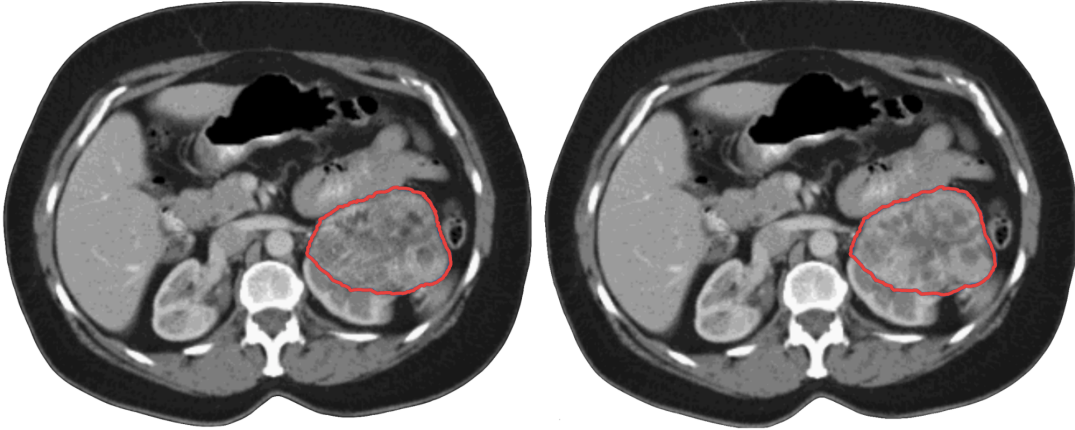


Figure 9. Example of a CT image containing a tumor generated by MAISI after fine-tuning for 10,000 epochs on a single scan (left), compared with the same real tumor-containing CT scan (right). The tumor region is outlined in red in both images.

To make this visual assessment possible during training, we additionally implemented a custom function that generated, decoded, and saved one sample at fixed epoch intervals. This enabled us to monitor how the quality of the generated outputs evolved over time and to stop training early if the sample quality did not improve. After finalizing the fine-tuning of the ControlNet, we turned to the next stage of the work, namely adapting the MAISI framework for inpainting.

3.2.3 Adapting MAISI for inpainting

The original MAISI implementation was designed to generate an entire CT volume rather than modify only a selected region. In our setting, however, the goal was not to synthesize a complete scan from scratch, but to insert a realistic kidney tumor into an existing control CT scan. This can be formulated as an inpainting task: the model is given the original image together with a predefined tumor mask, and it generates new image content only inside the masked region while preserving the surrounding anatomy unchanged. This was particularly important because

the resulting synthetic scans were later used in downstream segmentation experiments, where a realistic anatomical context was needed. For this reason, we first examined how the original MAISI generation pipeline operated and then investigated which modifications were required to adapt it for inpainting.

After analyzing the original pipeline, we introduced a new input to the sampling procedure: the latent embedding of a healthy scan into which a tumor was to be generated according to a provided segmentation mask. Instead of initializing the latent representation entirely with random noise, the healthy embedding was loaded and noise was added only within the masked tumor region. In this way, the model generated new content only where the tumor should appear, while the rest of the scan remained tied to the original anatomy.

We explored two related inpainting strategies. In the first approach, the diffusion process was run on the partially noised latent representation, and after all denoising timesteps were completed, the generated tumor region was inserted into the original healthy embedding. The autoencoder then reconstructed the final 3D CT image from this combined latent representation. In the second approach, inspired by the work of Jiang et al. [71], the tumor region was updated at every diffusion timestep using the output of the frozen latent diffusion model, while the regions outside the tumor mask remained unchanged throughout the process. After the final timestep, the autoencoder reconstructed the full 3D image containing the generated kidney tumor.

Based on visual evaluation, the second approach was selected for generating kidney tumors on TUH control scans for the subsequent experiments, as it produced more consistent and visually plausible results. Once this inpainting strategy was established, we also investigated alternative synthetic data generation methods to assess how the modified MAISI framework performed in comparison with other synthetic data generation approaches.

3.2.4 Alternative synthetic data generation methods

For the downstream segmentation experiments, we selected four alternative synthetic data generation methods: the 2D method, the 2.5D method, DiffTumor, and TS-Radiomics. The 2D and 2.5D methods are already described in Section 2.4 and are therefore not discussed further here, since the corresponding generations were produced by the co-supervisors and subsequently used in this work. In contrast, DiffTumor and TS-Radiomics were not directly applicable to our task and therefore required adjustments to the framework. However, because of limitations of the TS-Radiomics framework, we were not able to use it for final synthetic tumor generation, even though we attempted to adapt it to our task.

Based on our inspection of the framework and preliminary implementation experiments, we found that the publicly available TS-Radiomics framework⁴ was primarily designed for brain tumor synthesis. Therefore, its preprocessing pipeline had to be adapted for kidney CT data, particularly with regard to the intensity scaling range, which we refer to here as windowing. Since the framework did not provide a script for radiomics feature extraction, even though these features were essential for tumor generation, this step was implemented separately and tested on a small number of scans. In addition, the authors did not provide pretrained checkpoints for the models included in the framework, which meant that both the autoencoder and the latent

⁴ <https://github.com/jongdory/TS-Radiomics>

diffusion model had to be trained from scratch. We therefore first attempted to train these components on a few scans in order to verify that the pipeline worked correctly.

However, when testing image compression on a TUH control scan with the same size as in the MAISI experiments, $512 \times 512 \times 512$, we encountered an out-of-memory error even on a single H200 GPU with 141 GB of memory. Further investigation showed that TS-Radiomics did not include memory-saving mechanisms comparable to those used in MAISI, such as volume splitting during autoencoding. As a result, the largest image size that could be processed reliably was $128 \times 128 \times 128$, which was too small for the downstream segmentation task and did not allow for a fair comparison with the other approaches. For this reason, we did not proceed further with this framework and instead turned to another approach, namely DiffTumor.

Our inspection of the framework and the preliminary implementation experiments showed that the original DiffTumor⁵ pipeline was designed primarily for augmenting segmentation model training rather than for generating standalone synthetic datasets. It randomly sampled tumor locations and shapes within user-provided healthy CT scans and generated synthetic tumors on-the-fly during segmentation training, without saving the generated images as independent outputs. In our work, however, the goal was to explicitly control the location and shape of generated kidney tumors by providing both healthy CT scans and predefined tumor masks. To achieve this, we implemented an additional inference function that accepted a healthy CT volume, a user-defined tumor mask, and generated tumor appearance within the specified region. Since the diffusion model operates on fixed-size $96 \times 96 \times 96$ patches, we first cropped a region around the provided tumor mask, normalized the cropped CT patch to the intensity range used during DiffTumor training, performed tumor synthesis on the cropped region, converted the generated output back to HU windowed values $[-160, 240]$, and finally pasted the generated patch back into the original full-resolution CT volume. This modification made it possible to save complete synthetic CT scans and corresponding labels as standalone files, which the original DiffTumor pipeline did not support by default.

In addition to modifying the DiffTumor implementation, we also attempted to fine-tune the pretrained checkpoint of the latent diffusion model on our dataset of 850 CT scans. However, after multiple attempts at both fine-tuning and training from scratch, we ultimately chose to use the original pretrained checkpoint for kidney tumor generation for all TUH control scans as the generation quality degraded when training was continued on our data, suggesting that the model gradually forgot part of its prior knowledge as training progressed. With the final generation setups established, we then proceeded with the creation of the synthetic datasets used in the downstream segmentation experiments.

3.3 Synthetic data generation

As already mentioned, all synthetic datasets were generated from the same 160 control scans and their corresponding kidney tumor masks. Using these common inputs, we generated synthetic kidney tumors with our modified MAISI framework and with DiffTumor, while the datasets produced using the 2D and 2.5D methods were provided by the co-supervisor. This common input setup ensured that the main differences between the resulting datasets were due to the generation method itself rather than to differences in the underlying anatomy.

⁵ <https://github.com/MrGiovanni/DiffTumor>

Before generation, the control scans and masks were preprocessed in a way compatible with the modified MAISI framework. Specifically, the scans were cropped or padded around the center to a size of $512 \times 512 \times 512$, so that the kidneys and the generated tumor region remained fully visible. This was important because the same scans were later used in the downstream segmentation experiments, where incomplete anatomy could negatively affect model performance. In contrast, this requirement was less strict during MAISI fine-tuning, where the main objective was only to enable the model to learn kidney tumor generation. The selected size also had to be compatible with the MAISI framework, which supports only certain image sizes. In addition, the voxel spacing was standardized to $1 \times 1 \times 1$ mm and the orientation was converted to RAS. A copy of each segmentation mask was then created in which only the tumor region was retained, and its label was changed to match the MAISI label convention. Finally, the CT intensity values were clipped to the range -1000 to 1000 HU, as this was the input range expected by the MAISI framework. For DiffTumor, the same preprocessed scans served as the basis for generation, but the CT intensities were additionally windowed to the range $[-160, 240]$, and the masks followed the original label convention rather than the MAISI-specific one.

As a result, four synthetic datasets were obtained, one for each generation method, all based on the same control scans and tumor masks but differing in the appearance and quality of the generated tumors. These datasets were then used to assess how the choice of generation method influences segmentation performance.

3.4 Segmentation experiments

The main purpose of the downstream segmentation experiments was to evaluate how synthetic data generated by different approaches affect segmentation model performance, and how this performance changes when gradually increasing proportions of real data is added. As in the synthetic data generation pipeline, the segmentation experiments also began with data preprocessing.

3.4.1 Preparation of training and test sets

For the downstream segmentation experiments, both synthetic and real training data was used. Since each synthetic dataset contained 160 scans, 160 real tumor-containing CT scans provided by Better Medicine were also selected for training in order to keep the training set sizes comparable.

The remaining 31 tumor-containing scans from the TUH dataset were reserved for testing. In addition, the 30 control scans with corresponding segmentation masks that remained outside the subset used for synthetic data generation were also included in the test set. As a result, the final test set consisted of 61 CT scans with ground-truth segmentation masks. This setup allowed the segmentation models to be evaluated on both tumor-containing and healthy scans, making it possible to assess not only tumor detection performance but also the occurrence of false positive predictions.

As mentioned previously, the synthetic data generated using the 2D and 2.5D methods were provided by the co-supervisor and had already been windowed, meaning that the HU values were clipped to the range $[-160, 240]$. To ensure a fair comparison across generation methods, the synthetic data produced by the other approaches were also windowed in the same way. In particular, the scans generated by our modified MAISI, whose original HU values ranged from -1000 to 1000 , were clipped to the same interval. In contrast, the data generated by DiffTumor did not require additional windowing, as the framework already produced scans within the HU range specified in its configuration, in our case, to the range $[-160, 240]$.

Additional preprocessing was required for the data generated by the 2D and 2.5D methods because the scan sizes varied. To make these data comparable with the scans generated by our modified MAISI, they were similarly cropped or padded around the center of the scan to a size of $512 \times 512 \times 512$. Voxel spacing and orientation had already been standardized to the expected values by the co-supervisor before the data was provided.

Since real data was also used in the segmentation experiments, these scans were standardized in the same way. Specifically, they were cropped or padded to a size of $512 \times 512 \times 512$, the voxel spacing was standardized to $1 \times 1 \times 1$ mm, the orientation was converted to RAS, and HU values clipped to the range $[-160, 240]$. Without such standardization, differences in image size, spacing, orientation, and HU values could negatively affect segmentation model performance and complicate the interpretation of the results. After these preprocessing steps were completed, the data was ready to be used for training the segmentation models.

3.4.2 Training details

We ran the same set of downstream segmentation experiments for each of the four synthetic data generation methods under several training data settings. To establish a baseline, we first trained the segmentation model using only 160 real tumor-containing scans from the TUH dataset. We then trained the model on the synthetic-only dataset and on combinations of the full synthetic dataset with increasing proportions of real data. Specifically, we gradually added 5%, 10%, 25%, 50%, 75%, and 100% of the real training data in separate runs, corresponding to 8, 16, 40, 80, 120, and 160 scans, respectively. When moving from one proportion of real data to the next, the previously selected real scans were retained and only additional scans were added. In other words, the real-data subsets were cumulative rather than independently resampled for each setting. This design reduced the chance that the results would be influenced by which particular scans happened to be selected. We chose to keep the synthetic dataset fixed and gradually increase the amount of real data because the main aim of this work was to evaluate whether synthetic data can reduce the need for large amounts of annotated real data. This setup therefore reflects the intended use case, namely segmentation training in settings where real data is limited.

For the segmentation experiments, we used nnU-Net [9], a self-configuring framework that automatically adapts preprocessing, network configuration, and training settings to the properties of the input data, thereby reducing the need for manual design choices. This is one of the main reasons why we selected nnU-Net for our experiments: instead of manually tuning hyperparameters, we could directly run the training pipeline and let the framework determine the most suitable settings automatically. By default, nnU-Net uses 5-fold cross-validation, meaning that the training set is divided into five folds and, in each run, one fold specified by the user is used for validation while the others are used for training. The validation fold is used to evaluate the model on unseen data during training, providing an estimate of its generalization performance. In our experiments, however, we used only three out of the five folds because of time constraints. Consequently, each dataset setting was trained and evaluated on the same test set three times, once for each selected fold. This was done to reduce the influence of a particular train-validation split on the final results. Since performance can vary across folds, evaluating multiple folds and averaging the results provides a more reliable estimate of the segmentation model’s performance. We also observed that, for our setting, evaluating the results on three folds was sufficient. After completing training and evaluation on the test set, we calculated the evaluation metrics described in the following subsection for each method and dataset setting.

3.4.3 Evaluation metrics

In this work, we analyze four evaluation metrics: Dice score, false positive rate, sensitivity, and specificity. Among these, the Dice score is computed at the voxel level, whereas false positive rate, sensitivity, and specificity are computed at the scan level. This combination of metrics is chosen because it captures both segmentation quality in terms of tumor-mask overlap and the model’s ability to detect tumors while avoiding false positive predictions.

The Dice score used in this work is defined as follows:

$$Dice = \frac{2|P \cap G|}{|P| + |G|},$$

where P denotes the set of voxels predicted as tumor and G the set of ground-truth tumor voxels.

The voxel-level Dice score is suitable in this work because it directly measures the overlap between the predicted tumor region and the ground-truth tumor region. It is therefore informative for evaluating how accurately the segmentation model delineates tumors. Its values range from 0 to 1, where 1 indicates perfect overlap between the prediction and the ground truth.

The next metric we analyze is the false positive rate. In this work, it is calculated as the proportion of scans that contain at least one incorrectly predicted tumor region, with lower values indicating better performance. Unlike the Dice score, which measures voxel-level overlap between predicted and ground-truth tumor regions, this metric is evaluated at the scan level over the entire test set, including both tumor-containing and control scans. It is therefore useful for assessing how well the model avoids false tumor predictions more generally.

Because the Dice score reflects voxel-level segmentation quality rather than scan-level tumor detection, and the false positive rate alone does not fully capture how well the model separates tumor-containing from non-tumor-containing scans, we additionally analyze sensitivity and specificity. In this work, sensitivity reflects the proportion of tumor-containing scans that are correctly identified by the segmentation model, with higher values indicating better performance. A scan is counted as a true positive if the predicted tumor region overlaps sufficiently with the ground truth, where sufficient overlap is defined as an intersection over union (IoU) of at least 0.3. If this condition is not met, the scan is counted as a false negative. Sensitivity is therefore particularly informative for evaluating how reliably the model detects tumors. Sensitivity is defined as follows:

$$Sensitivity = \frac{TP}{TP + FN},$$

where TP denotes the number of true positive scans and FN the number of false negative scans.

Specificity, on the other hand, reflects the proportion of non-tumor-containing scans that are correctly identified by the segmentation model. In other words, it measures how well the model avoids false tumor predictions on healthy scans, with higher values indicating better performance. Specificity is defined as follows:

$$Specificity = \frac{TN}{TN + FP},$$

where TN denotes the number of true negative scans and FP the number of false positive scans.

Having described the datasets, synthetic data generation methods, downstream segmentation setup, and evaluation metrics, we can now proceed to the analysis of the segmentation results across the different methods and dataset settings.

4. Results

In this section, we present the results of the segmentation experiments conducted to evaluate the usefulness of synthetically generated CT data for kidney tumor segmentation. The comparison includes different synthetic data generation methods and multiple training settings, ranging from synthetic-only data to mixtures of synthetic and real data. We first evaluate the generated tumors through visual inspection and then assess the performance of the segmentation models using tumor Dice score, false positive rate, sensitivity and specificity. Together, these metrics provide insight into both segmentation quality and the trade-off between correctly detecting tumor-containing scans and avoiding false positive predictions.

4.1 Visual assessment of synthesized tumors

Before starting the segmentation experiments, we first evaluated the generated tumors through visual inspection by selecting several samples from each synthetic dataset. For visualization of the 3D volumes, we used 3D Slicer⁶, an open-source software platform for the visualization, processing, segmentation, registration, and analysis of medical and biomedical 3D images. In the visual assessment, we mainly focused on the tumor texture, intensity, and how well the generated tumor filled the corresponding segmentation mask. We also examined whether the tumor blended naturally with the surrounding kidney tissue and whether any visible artifacts or intensity mismatches were present. This helped us assess whether the synthetic tumors were visually plausible before using them in the segmentation experiments.

The visual comparison of the different generation methods is illustrated in Figure 10. We observed that, although all methods were capable of generating tumors within the provided masks, the visual quality varied considerably. For example, DiffTumor tended to produce unrealistically dark tumors, while the 2D approach occasionally generated very bright tumors. Based on our observations, among these approaches, 2.5D and our modified MAISI produced tumors that were closest to the appearance of real tumors, with plausible texture, contrast, and intensity. As mentioned, we also paid attention to how well these approaches filled the segmentation masks. Our visual inspection showed that all methods mostly filled the masks well; however, we noticed that generated tumors occasionally appeared faded near the boundaries across all methods. This is likely related to the difficulty of modeling the transition between the synthesized tumor and the original surrounding tissue. Since the methods aim to preserve anatomical consistency outside the tumor region, the generated boundaries may become overly smooth, which reduces contrast and makes the tumor margins appear faded.

Since CT volumes are three-dimensional, we also inspected the consistency of the synthetic tumors across consecutive slices. Figure 11 shows tumor crops from five consecutive slices for each approach. As shown in the figure, tumors generated using the 2D method are not always consistent from slice to slice. This is most likely because, in this method, the model generates a tumor for each slice independently, without using information from adjacent slices. Such inconsistency may make it more difficult for the downstream segmentation model to learn realistic tumor patterns, since real tumors usually show spatial continuity across neighboring slices. In contrast to the 2D method, the other approaches maintained better consistency across

⁶ <https://www.slicer.org/>

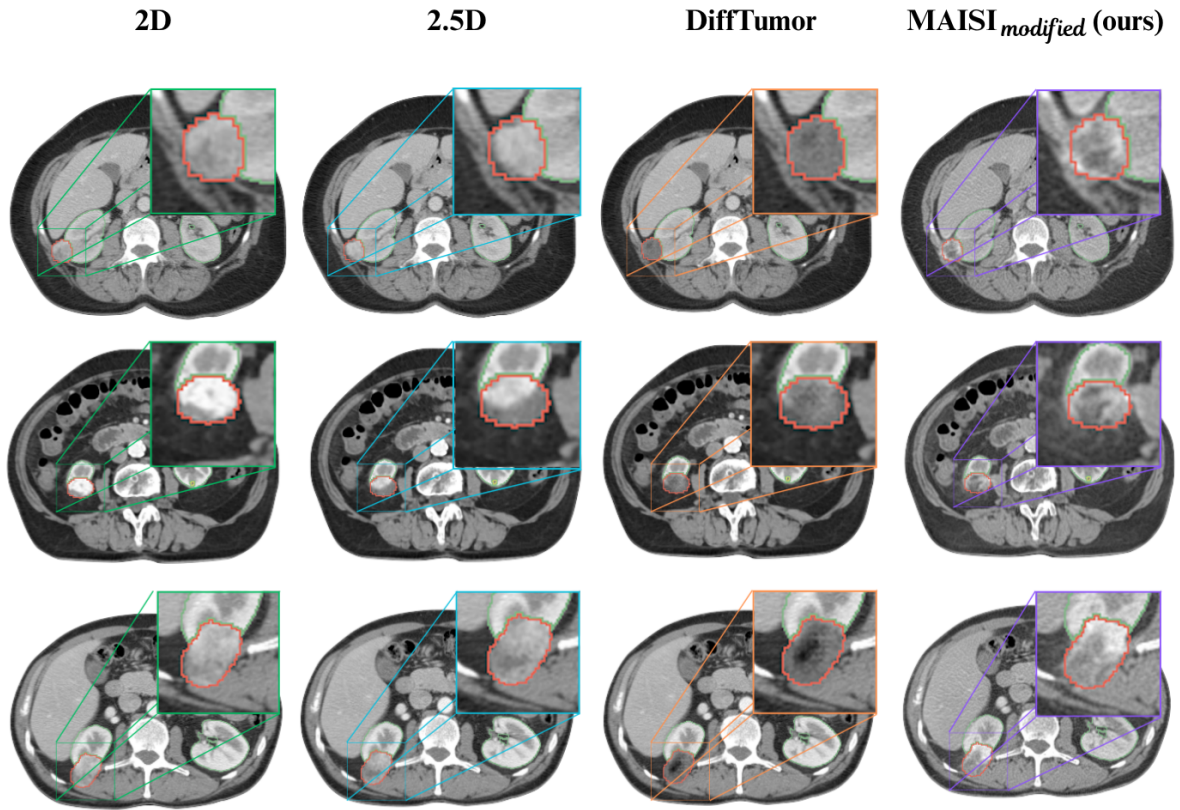


Figure 10. Visual comparison of synthetic tumor generation methods. Each row represents a different case, while each column corresponds to a different generation method: 2D, 2.5D, DiffTumor, and our modified MAISI. The zoomed-in regions highlight tumor appearance.

slices, as they were adapted for 3D data or, in the case of the 2.5D method, incorporated information from adjacent slices.

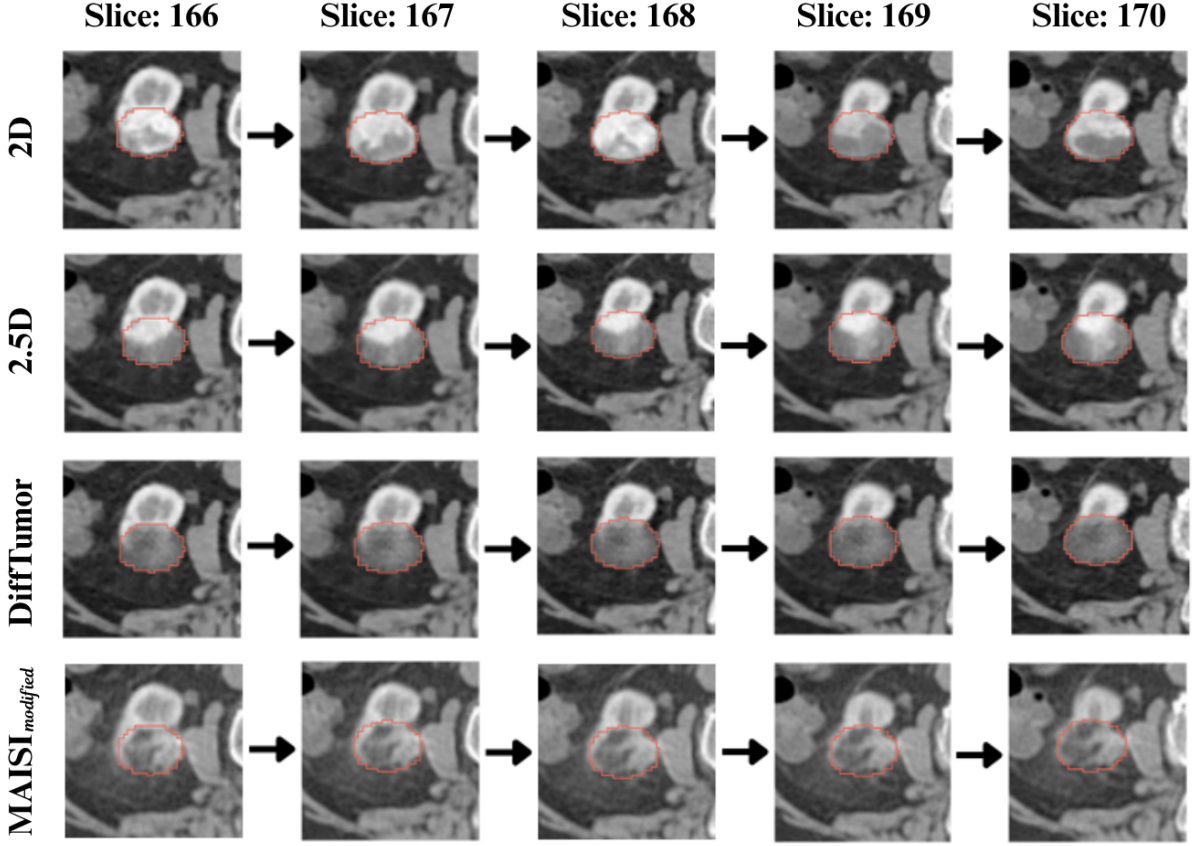


Figure 11. Tumor crops from five consecutive slices generated using four methods: 2D, 2.5D, DiffTumor, and our modified MAISI. The figure highlights how consistently the generated tumors are represented across adjacent slices. For the 2D method, the tumor texture and contrast change noticeably from one slice to the next, whereas the other methods show a more consistent appearance across the volume.

However, visual inspection alone is not sufficient to draw conclusions about the quality of synthetic data, as it may contain artifacts that are not easily visible to the human eye but can still affect the performance of downstream segmentation models. Therefore, it is also important to analyze quantitative metrics obtained from the segmentation experiments, such as Dice score, false positive rate, specificity, and sensitivity.

4.2 Analysis of segmentation performance

In this section, we discuss the segmentation results obtained for the different generation methods and dataset settings. The complete results for all evaluated metrics are provided in the Table 1.

The first metric we analyze is the Dice score, which in this work is calculated at the voxel level and ranges from 0 to 1, where 1 indicates perfect overlap between the predicted and ground-truth tumor masks. We use the score calculated using only real data as the baseline, with a value of 0.86.

As shown in Table 1, our modified MAISI achieves the highest Dice score in 4 out of 7 dataset settings when compared with the other generation methods. The largest performance gap is

Table 1. Summary of segmentation performance across different training datasets and methods. The reported values are averaged across the three folds. The baseline corresponds to the model trained using only real data.

Method	Synth only	Synth + 5% real	Synth + 10% real	Synth + 25% real	Synth + 50% real	Synth + 75% real	Synth + 100% real	Real only baseline
Tumor Dice score (higher is better)								0.86
2D	0.02	0.51	0.67	0.74	0.83	0.83	0.83	
2.5D	0.06	0.60	0.73	0.81	0.83	0.85	0.86	
DiffTumor	0.03	0.56	0.69	0.81	0.83	0.84	0.87	
MAISI _{modified} (ours)	0.25	0.61	0.68	0.80	0.86	0.87	0.85	
False positive rate (%) (lower is better)								37.2
2D	9.8	61.2	35.0	38.8	27.3	32.2	38.3	
2.5D	26.2	30.1	36.6	36.1	43.7	32.2	43.7	
DiffTumor	21.3	38.8	40.4	39.9	38.8	35.0	42.1	
MAISI _{modified} (ours)	10.9	16.9	20.8	25.7	27.3	28.4	47.0	
Sensitivity (higher is better)								0.95
2D	0.01	0.61	0.76	0.84	0.91	0.91	0.90	
2.5D	0.06	0.71	0.83	0.90	0.92	0.94	0.95	
DiffTumor	0.02	0.62	0.80	0.92	0.92	0.92	0.94	
MAISI _{modified} (ours)	0.26	0.70	0.80	0.89	0.92	0.95	0.95	
Specificity (higher is better)								0.62
2D	0.92	0.43	0.73	0.71	0.79	0.71	0.66	
2.5D	0.72	0.68	0.73	0.67	0.60	0.69	0.58	
DiffTumor	0.81	0.70	0.69	0.67	0.66	0.70	0.61	
MAISI _{modified} (ours)	0.92	0.86	0.80	0.74	0.74	0.77	0.57	

observed in the synthetic-only setting, where our modified MAISI clearly outperforms the other methods, and this advantage remains visible when 5% real data is added. Overall, the results show that, regardless of the generation method, segmentation performance generally improves as the proportion of real data increases. This suggests that synthetic data alone is not sufficient for accurate delineation of tumor regions and that the addition of real data substantially improves the model’s ability to identify the tumor more precisely. Notably, the 2D method achieves the lowest performance in most dataset settings, which may indicate that consistency across slices is an important factor for segmentation performance. Among the selected methods, the 2D approach is the only one that generates tumors slice by slice without explicitly using information from adjacent slices. In addition, its Dice score improves only up to the setting with 50% real data, after which no further improvement is observed. This may suggest that the usefulness of the synthetic data generated by the 2D method becomes limited once a moderate amount of real data is available. Based on the Dice score, the modified MAISI reaches performance comparable to the real-data baseline already when 50% of real data is added, whereas the other methods (2.5D and DiffTumor) achieve their best results only when 100% real data is included.

However, the Dice score alone is not sufficient for determining which generation method performs best, because a model may achieve reasonable overlap with the ground truth while still producing false positive tumor predictions, which can be clinically problematic. False positive predictions may lead to unnecessary follow-up examinations, increased workload for clinicians, and reduced trust in the model’s outputs. Therefore, we also calculate the false positive rate to assess how often the segmentation models produce incorrectly predicted tumor regions.

In this work, the false positive rate is measured at the scan level as the proportion of scans that contain at least one falsely predicted tumor region, with lower values indicating better performance. The false positive rate obtained using only real data is used as the baseline and equals 37.2%.

The lowest false positive rates are generally achieved by our modified MAISI, particularly in the mixed-data settings with 5%, 10%, 25%, and 75% real data. This suggests that the synthetic data generated by the modified MAISI help the segmentation model avoid false tumor predictions more effectively than the other generation methods. At the same time, the 2D method achieves the lowest false positive rate (9.8%) in the synthetic-only setting and remains competitive at several mixing ratios, indicating that its synthetic data may be particularly effective for reducing false positives even if its Dice scores are lower in some cases. In contrast, the 2.5D and DiffTumor methods generally result in higher false positive rates, suggesting that the corresponding synthetic data may introduce patterns that make the segmentation models more prone to incorrect tumor predictions. Interestingly, for our modified MAISI, the false positive rate tends to increase as more real data is added, with the highest value observed when all available real data is used. This indicates that the relationship between synthetic and real data is not straightforward and that adding more real data does not necessarily improve all aspects of segmentation performance. To complement the false positive rate analysis, the Figure 12 and Figure 13 in Appendix also provide plots of the mean number of false positive tumor objects and the mean number of false positive tumor voxels across the evaluated methods and dataset settings.

Although the Dice score and false positive rate provide valuable information, they are not sufficient on their own. The Dice score does not directly indicate how many tumor-containing scans are correctly detected, and the false positive rate does not fully capture the balance between detecting tumors and avoiding false alarms. For this reason, we also calculate sensitivity and specificity.

Sensitivity measures the proportion of correctly detected tumor-containing scans. Sensitivity ranges from 0 to 1, with higher values indicating better performance. In this work, a scan is considered a true positive if the predicted tumor region overlaps sufficiently with the ground truth, defined as an intersection over union (IoU) of at least 0.3. This threshold was chosen to ensure that the prediction captures a substantial part of the tumor, while allowing for some boundary mismatch. The sensitivity obtained using only real data is used as the baseline, with a value of 0.95.

Based on the results, our modified MAISI and the 2.5D method perform similarly, achieving nearly identical sensitivity values across most datasets. Generally, sensitivity increases with the addition of real data, suggesting that real samples contribute substantially to reliable tumor detection and improve the ability of the models to correctly identify tumor-containing scans. Although the modified MAISI matches the baseline on two datasets and the 2.5D method on one, none of the methods surpasses the real-only baseline.

Another notable observation is that the differences between methods are largest in the synthetic-only setting, where the modified MAISI substantially outperforms the other methods. This suggests that the synthetic data generated by the modified MAISI is more effective for supporting reliable tumor detection, especially in the absence of real training data. Once real data is added, the sensitivity values of the best-performing methods (modified MAISI and 2.5D) become much closer, suggesting that real training data reduce the performance gap between the synthetic generation approaches. In contrast, the 2D method remains consistently below the

best-performing methods across all training datasets, indicating that its synthetic data is less effective for supporting robust tumor detection.

While sensitivity provides insight into how well the models correctly detect tumor-containing scans, we are also interested in how well they avoid predicting tumors on healthy scans. For this reason, we calculate specificity. Unlike the false positive rate, which is measured across the entire test set, including both healthy and tumor-containing scans, specificity focuses only on healthy scans and indicates how often false positive tumor predictions are correctly avoided.

In this work, specificity shows the proportion of correctly avoided false positive predictions across healthy scans, where values are in range from 0 to 1, with higher values indicating better performance. We used the specificity score obtained on real-only dataset as a baseline, with a value 0.62.

Based on the results, the best-performing model is our modified MAISI, which achieves substantially higher specificity across most datasets, highlighting its strong ability to avoid false positive predictions on non-tumor-containing scans. Interestingly, all methods exceed the real-only baseline on most datasets, suggesting that synthetic data may provide additional information that helps the models better distinguish healthy scans from tumor-containing ones.

Another notable observation is that the highest specificity values are generally achieved in settings with little or no real data. In particular, the modified MAISI and the 2D method both achieve the highest specificity (0.92) on the synthetic-only dataset. As larger amounts of real data is added, specificity tends to decrease for most methods, suggesting that the segmentation models become more likely to produce false positive tumor predictions on healthy scans. This trend is especially visible for the modified MAISI, whose specificity remains clearly above the baseline in most mixed-data settings, but drops below it when all available real data is used. These results are consistent with the false positive rate analysis, confirming that the modified MAISI is particularly effective at reducing false tumor predictions on healthy scans in most training settings.

Taken together, the results across all evaluated metrics and dataset settings suggest that our modified MAISI is the strongest method overall, although no single method is best in every setting. Its advantage is most pronounced when real data is scarce: it clearly outperforms the other methods in the synthetic-only setting and remains the strongest method when only 5% real data is added. As the proportion of real data increases, the differences between methods become smaller, and in some settings the 2.5D method or DiffTumor achieve comparable or slightly better values for individual metrics. Nevertheless, the modified MAISI shows the most consistent overall behavior across the evaluated metrics and reaches a particularly strong trade-off between segmentation accuracy, sensitivity, specificity, and false positive rate when combined with 75% real data. The results are summarized in Table 2.

4.3 Ablation study: tumor-based loss design

One possible reason for this stronger performance of the modified MAISI lies in the design of the loss function used during fine-tuning. In the original implementation, the loss is calculated over the entire scan, whereas in our approach it is calculated only over the tumor region and its surrounding area. The surrounding area is defined by expanding the tumor mask by five voxels in each spatial direction. Since all scans were standardized to spacing of $1 \times 1 \times 1$ mm, this corresponds to an approximately 5 mm margin around the tumor. This allows the optimization to focus more strongly on the image region most relevant to the target task while

Table 2. Segmentation performance in the synthetic + 75% real data setting across different synthetic data generation methods, with the real data baseline included in a separate column. The reported values are averaged across the three selected folds. For Dice score, sensitivity, and specificity, higher values indicate better performance, whereas for false positive rate, lower values indicate better performance.

Metric	2D	2.5D	DiffTumor	MAISI _{modified} (ours)	real data (baseline)
Dice score	0.83 $\pm$ 0.02	0.85 $\pm$ 0.01	0.84 $\pm$ 0.02	0.87 $\pm$ 0.02	0.86 $\pm$ 0.02
False positive rate	32.2 $\pm$ 9.03	32.2 $\pm$ 8.25	35.0 $\pm$ 2.5	28.4 $\pm$ 0.95	37.2 $\pm$ 0.95
Sensitivity	0.91 $\pm$ 0.02	0.94 $\pm$ 0.03	0.92 $\pm$ 0.02	0.95 $\pm$ 0.04	0.95 $\pm$ 0.02
Specificity	0.71 $\pm$ 0.1	0.69 $\pm$ 0.07	0.70 $\pm$ 0.03	0.77 $\pm$ 0.03	0.62 $\pm$ 0.04

still including the local anatomical context around the tumor. During development, we also explore a variant in which the loss is calculated only within the tumor region. However, visual assessment of the synthesized tumors shows that generation quality improves substantially when the surrounding area is included as well. This suggests that the model benefits from learning not only tumor-specific characteristics but also the local tissue context. Overall, these observations complement the quantitative results presented above and help explain why the final modified MAISI framework achieves stronger performance than the alternative approaches.

After analyzing the visual appearance of the generated kidney tumors and the quantitative evaluation metrics, we now proceed to discuss the main findings, the limitations of this work, and potential directions for future research.

5. Discussion

This thesis explored whether synthetically generated kidney tumor CT images can reduce the amount of annotated real-world data needed to train a segmentation model while maintaining competitive segmentation performance. The results show that, across the selected methods and dataset settings, the best trade-off between segmentation quality and avoidance of false positive predictions is achieved by combining synthetic data generated with our modified MAISI and 75% real data. In this setting, the reduction in required real data is moderate, namely 25%, but the overall performance is stronger than when training on real data alone. More specifically, the segmentation overlap is comparable to the real-only baseline, while the number of false positive predictions is reduced by approximately 9%.

Overall, our modified MAISI achieves better performance than the other generation methods in most dataset settings. The performance gap is especially noticeable when little or no real data is available. Although the ability to identify kidney tumors in these settings remains clearly weaker than when training on real data alone, and the ability to avoid false positive predictions is substantially stronger when only synthetic data, or synthetic data with a small proportion of real data, is used. In addition, the modified MAISI shows relatively stable results across folds, with variability comparable to that of the real-data baseline, whereas the false positive rate of the other methods tends to have a larger standard deviation, indicating less stable performance across folds.

One possible reason why MAISI performs better than the other methods is that it includes foundation models trained on a substantial amount of medical imaging data, allowing the framework to learn more general CT image patterns before being adapted to the kidney tumor task. Fine-tuning on our dataset then provides the framework with task-specific information about kidney tumors. In contrast, the other methods rely mainly on training with smaller amounts of data more narrowly focused on kidney tumors. In addition, the loss function modification introduced in our modified MAISI likely also contributes to the improved performance, since the model learns not only the tumor region itself but also its surrounding context, which may be especially important for generating realistic tumor boundaries.

At the same time, the results show that no single approach is best in every setting. The overall pattern suggests that adding more real data generally improves segmentation performance, but often also increases the false positive rate. The best trade-off appears to lie around 50–75% real data not only for MAISI, but also for the other generation methods, even though their highest segmentation performance is often reached only when 100% real data is included, typically at the cost of more false positive predictions. The 2D method, however, performs worse than the other methods in most settings. A likely reason is its architectural limitation, namely generating tumors slice by slice without explicitly using information from neighboring slices, which reduce consistency across the 3D volume.

Although the modified MAISI is capable of producing visually plausible tumors and achieves the strongest results in many dataset settings, segmentation performance when training only on synthetic data remains substantially lower than the real-data baseline. This indicates that synthetic data alone is not sufficient for training a fully reliable segmentation model. Instead, the results suggest that synthetic data is better understood as a practical supplement to real data rather than as a complete replacement.

While these findings highlight the potential of synthetic data, they also need to be interpreted in the context of several limitations of the present study. In the following section, we therefore discuss the main limitations of this work and outline possible directions for future research.

5.1 Limitations

This work has several limitations that should be taken into account when interpreting the results. First, MAISI is a large and complex framework, and not all possible combinations of hyperparameters were explored during ControlNet fine-tuning, mainly because of time and computational constraints. As a result, the full potential of the framework may not have been fully assessed. Second, only a limited number of real–synthetic mixing strategies were evaluated in the downstream segmentation experiments. Exploring a broader range of combinations could provide a more complete picture of how synthetic and real data interact.

Another limitation is that one of the selected generation methods, TS-Radiomics, could not be included in the final segmentation experiments. Although we attempted to adapt the framework to our task, we were not able to process image sizes larger than 128^3 , even on an H200 GPU with 141 GB of memory. This appears to be due to the fact that the framework architecture is not designed to encode large input images into latent representations efficiently. As a result, the maximum feasible image size was too small for a fair comparison with the other approaches used in this work. Finally, all segmentation models were evaluated on scans from a single data source. Therefore, it remains uncertain how well the findings would generalize to data acquired at other hospitals or with different scanner settings.

5.2 Future work

Several directions for future work follow naturally from these limitations. One possibility is to explore the MAISI framework more extensively by testing a wider range of hyperparameter settings and training configurations. This could help determine whether the performance of the modified framework can be improved further. Future work could also investigate ways to adapt the TS-Radiomics pipeline for larger 3D medical images, which would make it possible to include this method in a more complete comparative evaluation.

Another promising direction is to improve the DiffTumor framework, particularly by exploring architectural or training modifications that could reduce the loss of previously learned knowledge during fine-tuning. More generally, future studies could extend the evaluation to datasets from multiple institutions in order to assess the robustness and generalizability of the findings across different clinical settings.

6. Conclusion

In this work, we investigated whether synthetically generated data can reduce the amount of annotated real data needed to train a segmentation model while maintaining competitive performance. To address this question, we adapted the MAISI framework for kidney tumor generation by fine-tuning the ControlNet component and modifying the generation pipeline for inpainting, while alternative generation methods were used as comparative baselines. The results show that our modified MAISI outperforms the other methods in many dataset settings, especially when little or no real data is available. In general, segmentation performance tends to improve as more real data is added, but this often comes at the cost of an increased false positive rate. However, the strongest trade-off between segmentation quality and robustness to false positive predictions is achieved when synthetic data generated by the modified MAISI is combined with 75% real data. In this setting, segmentation performance is comparable to, or slightly better than, the real-only baseline, while the number of false positive predictions is reduced. At the same time, the experiments also show that synthetic data alone is not sufficient to match the performance achieved with real data.

Overall, these findings suggest that synthetic kidney tumor CT data can serve as a useful supplement to real annotated data, especially in low-data settings, but do not yet provide a full replacement for real training data. Among the methods considered in this work, the modified MAISI proves to be the most promising approach for generating synthetic data that is useful for downstream segmentation. These results provide a basis for further research on generative medical imaging methods and their potential to support model training in settings where annotated data is limited.

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Appendices

This section contains supplementary visualizations related to false positive predictions, including plots of the mean number of false positive tumor objects and the mean number of false positive tumor voxels across methods and dataset settings.

Mean number of false positive tumor objects

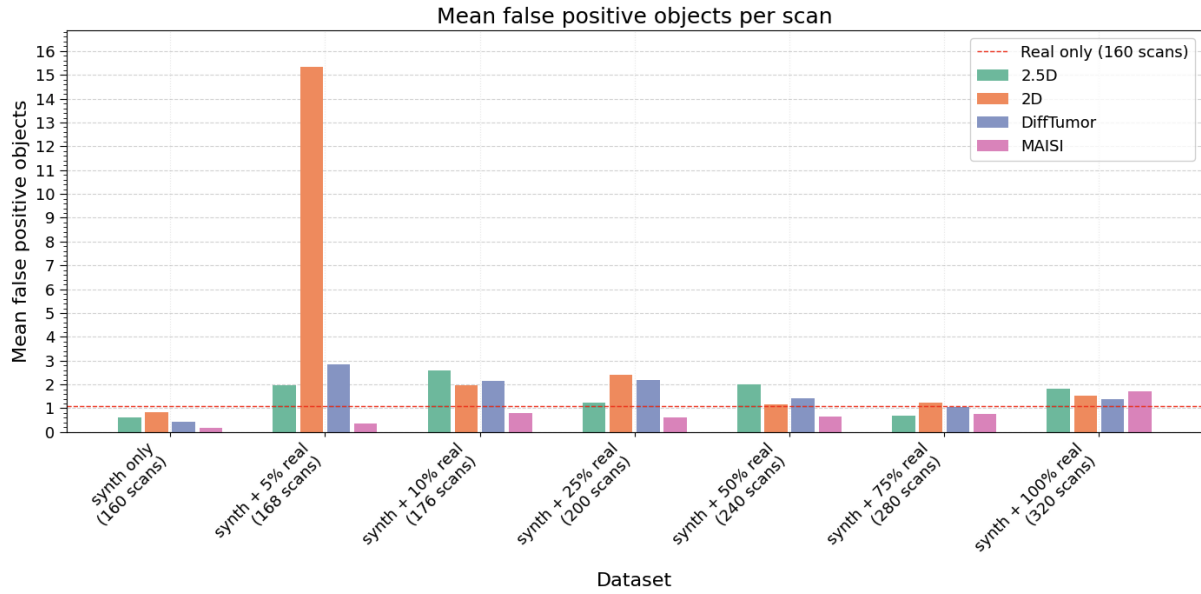


Figure 12. Mean number of false positive tumor objects per scan for the different synthetic data generation methods and training dataset settings. Lower values indicate that the segmentation model produces fewer spurious tumor objects. The dashed horizontal line represents the baseline obtained using only real data.

Mean number of false positive tumor voxels

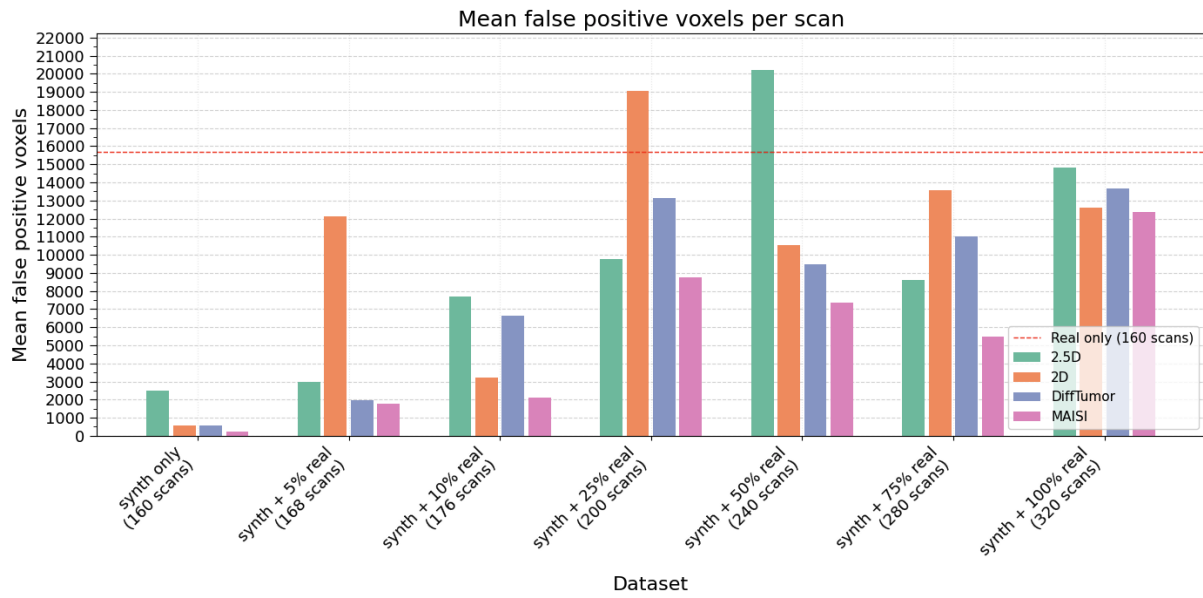


Figure 13. Mean number of false positive tumor voxels per scan for the different synthetic data generation methods and training dataset settings. Lower values indicate that the segmentation model produces fewer spurious tumor objects. The dashed horizontal line represents the baseline obtained using only real data.

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